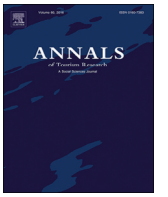
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RESEARCH NOTE

The use of synthetic data in tourism



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Introduction

Will a future reliant on synthetic data for tourism research lead to progress or stagnation? While synthetic data promises cost-effectiveness and accuracy in mimicking human behavior, its inability to capture evolving real-world behaviors raises concerns. Can we truly understand and study the dynamic nature of tourism without real-world data?

Large language models utilize vast text databases to predict the next word or part of it. These artificial intelligence systems are trained using language modeling and human feedback-based reinforcement learning (Achiam et al., 2023), enabling them to mimic human responses (Grossmann et al., 2023). While OpenAI's Generative Pre-trained Transformer (GPT) has been the primary focus in tourism and hospitality (Dwivedi et al., 2024), researchers have also examined other models like Gemini, Bard, and Claude 3 (Wu et al., 2023).

Researchers across various disciplines have started using large language models as substitutes for human participants in empirical studies (e.g., Argyle et al., 2023; Dillion et al., 2023; Li et al., 2024), including tourism (Sop & Kurçer, 2024). These models generate synthetic data that seek to mimic human responses to describe, explain, and predict human behavior. Deontological ethics may categorically oppose such synthetic data, arguing that they fundamentally differ from human respondent data, thus violating research values and responsibilities (Hutson & Mastin, 2023). Conversely, a utilitarian perspective would pragmatically consider the low costs and high benefits of using large language models for such predictions (Sarstedt et al., 2024). A key argument here is that large language models are a suitable means to generate large-scale data for specific research areas very quickly and at low costs (Li et al., 2024). But would synthetic data live up to this promise in tourism research?

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Issues and applications

A lingering question is understanding whether the data provided by large language models is similar to human data. Several studies in various fields, such as human-computer interaction, general psychology, and social psychology—albeit not in tourism research—offer synthetic and human data comparisons. These studies typically prompt a large language model with specific tasks or paradigms relevant to the respective field (e.g., vignette-based tasks) and compare the output with human responses (e.g., Binz & Schulz, 2023; Dillion et al., 2023; Hagendorff et al., 2023). Sarstedt et al.'s (2024) systematic review of research in this field discloses considerable variability in synthetic data's efficacy for mimicking human decision-making. In merely 99 (34.74 %) of the 285 comparisons, the synthetic datasets produce similar results as their human sample counterparts.

Whether or not synthetic data can mirror human responses largely depends on the training data (i.e., the corpus of data on which the large language model was originally trained) and its applicability to specific research areas. Many large language models use a reinforcement learning approach in which human annotators train the model to behave in a certain way. Additional fine-tuning (i.e., retraining the model on further, more specific data) seeks to adjust the large language model's parameters to fulfill a task better and avoid unwanted human biases (e.g., not perpetuating racist content).

Therefore, a large language model's output does not represent any genuine human response but rather a cross-section of numerous, heavily filtered, and pre-processed responses without reference to a clearly defined population (Fig. 1).

Researchers, therefore, need to critically evaluate how well the training data can address their research question (Santurkar et al., 2023). The accuracy of results from synthetic data may depend on the representation of the research area within the training data. For example, sustainable tourism research, often well-documented with data on carbon emissions, might yield more accurate synthetic data than research on a niche rural destination, which may need to be more represented in the training data. To address this issue, researchers may use retrieval-augmented generation, which is when the large language model is supplemented with documents or text passages that contain helpful information. However, standard large language models like OpenAI's GPT do not implement such retrieval-augmented generation capabilities.

Furthermore, large language models still need help to adequately capture nuanced personal differences that are central to tourism research (e.g., Adhikary & Adhikari, 2019; Park et al., 2024; Santurkar et al., 2023). In fact, large language models rarely associate information in textual data with identifiers (e.g., demographics), hampering possible inferences regarding a specific subgroup (Wang et al., 2024).

Researchers can improve the performance of large language models through prompt-tuning. This technique provides the model with sample tasks and corresponding solutions without adjusting the model parameters (Demszky et al., 2023). For instance, when investigating the relationship between flexwork and flextravel (see Miao et al., 2024), a researcher may prompt: "Here is an example of a flexwork-flextravel for a tourist: 'Imagine to be an individual leveraging flexwork arrangements for travel, for example, employees adhering to the Tuesday-to-Thursday Model or compressed workweeks.'" When tuning their prompts, researchers should be aware that their prompt design may reflect their prior expectations, a bias similar to *p*-hacking, which would limit the output's generalisability.

Considering the above-mentioned issues, using synthetic samples for substantial research phases such as main studies and large-scale quantitative research is challenging at best. This is because large language models often do not fully mimic human response behavior and do not capture the variance found in human samples (Park et al., 2024; Sarstedt et al., 2024). Researchers would need to validate results from the synthetic data with a human sample or another data source.

However, large language models show promise as an efficient solution for enhancing early-stage research processes regarding data generation speed and quality as long as researchers can validate and fact-check the outputs. The results may guide follow-up empirical analyses. For example, researchers can leverage large language models for scale pretesting to evaluate the adequacy of item wording. In Sarstedt et al. (2024), GPT-4 identified the item "I use this service very often" as unspecific, highlighting that the interpretation of "very often" may vary among individuals. Hoffmann et al. (2024) show that ChatGPT-3.5 aids scale development processes if its outputs are critically evaluated by experts in the field.

Large language models can also help to examine stimuli and identify confounding variables. Following Liu et al.'s (2024) work on the impact of resident pro-environmental behavior on tourists, we used stimuli from their Study 2b (see Fig. 2) to assess the

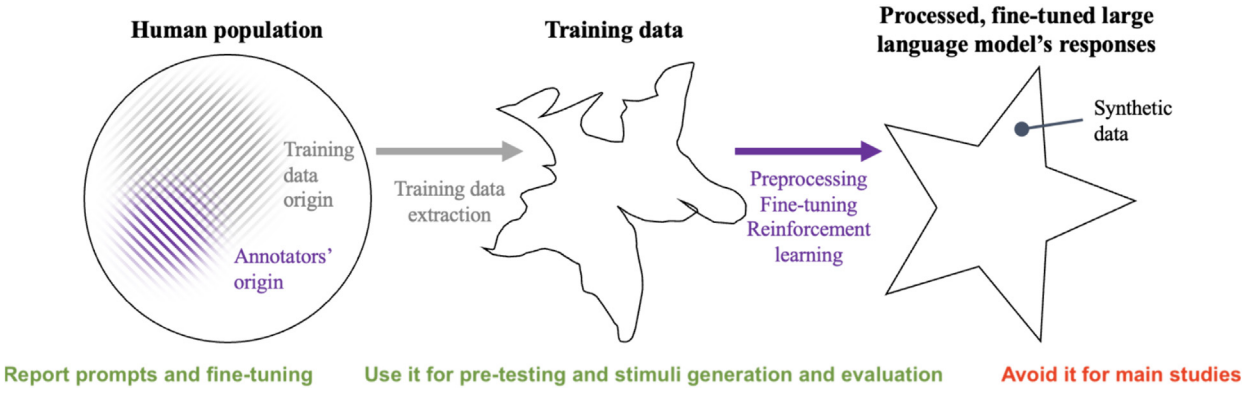


Fig. 1. Who do synthetic data represent and in which context can they be beneficial?



Fig. 2. Stimuli used in Study 2b by Liu et al. (2024).

appropriateness of generative AI tools like DALL-E and Gemini for evaluating visual stimuli. Specifically, we asked Gemini-1.5_Pro-128k to assess whether—as meant by the authors—the bag on the left seems recycled and the one on the right a plastic bag. The language model agrees but suggests a possible modification to avoid a confounding factor. Specifically, the bag on the left seems made of paper instead of plastic. Therefore, the large language model suggests comparing a recycled paper bag to a plastic bag or comparing a paper bag to a plastic bag (see the Appendix). Tourism researchers could further leverage this image identification feature to examine shots from actual touristic experiences (e.g., hotel or beach pictures).

The way forward

Synthetic data can offer a cost-effective and time-efficient solution for tourism research. They promote equity by enabling researchers with limited funding to conduct studies they might not otherwise afford.

In the long run, researchers could use large language models to interrogate textual data. For example, much knowledge that is relevant to tourism research (e.g., destination-specific reviews or travel diaries) is already stored in the training data. Researchers may also chat with a large specific corpus of text data (e.g., traveler reviews or destination descriptions) when assisted by retrieval-augmented generation. However, large language models will likely misinform tourism researchers if adopted in

Table 1
Checklist for using synthetic data in tourism.

Recommendation	Recommended reading
Use large language models for early stages in the research process (e.g., for pre-testing, or for stimuli generation and evaluation) and safeguard the output's validity using expert judgment or pilot studies	Sarstedt et al. (2024)
Apart from human-synthetic data comparisons, avoid using large language models for main studies.	
Reflect on the contents of the training data: For which research questions can they be informative?	Wang et al. (2024)
Do not use synthetic data when researching highly specific subpopulations or very recent developments, which are underrepresented in the training data. Caution should be executed when using synthetic data to study individual differences. Consider using retrieval-augmented generation, if suitable data is available.	Wang et al. (2024)
Always be transparent on the specific large language model type and version, the prompts (including prompt-tuning activities), and the parameter settings used (e.g., temperature). Report any fine-tuning used.	Demszky et al. (2023)
Ensure explainability and accountability of the model, its training data, the sample, and the outputs obtained. Preferably rely on open-source Large language models.	Bail (2024)

situations that are unlikely to be adequately captured in the training data or in additionally provided data. This could have severe consequences for the field, which is already under scrutiny due to its lack of real impact (Tribe & Paddison, 2024).

We therefore recommend caution when using large language models in main studies. Currently, insights from large language models should always be triangulated with other data sources—for example, additional quantitative data or qualitative interviews. Identifying when and how to use large language models as a stand-alone data source for main studies is a major undertaking that requires validation studies and academic discussion within and beyond tourism research. Contemporarily, the large language models' particular strength rather lies in the early stages of the research process, where researchers can check the output's adequacy using expert judgment or pilot studies.

A lingering concern regarding the use of large language models pertains to their probabilistic nature and strong reliance on fine-tuning, which hinder reproducibility and replicability (Bail, 2024). Tourism researchers should, therefore, be transparent in their parameter settings (e.g., concerning temperature settings) and preferably rely on open-source large language models to avoid the black box character of proprietary models (Spirling, 2023).

Summarizing our discussion above, Table 1 offers a series of recommendations for tourism research.

CRediT authorship contribution statement

Giampaolo Viglia: Writing – original draft. **Susanne J. Adler:** Visualization, Validation. **Caroline Lancelot Miltgen:** Supervision, Investigation. **Marko Sarstedt:** Methodology, Conceptualization.

Declaration of competing interest

The authors have no relevant financial or non-financial interests to disclose. The authors have no conflicts of interest to declare relevant to this article's content.

Appendix. Prompt and response from Gemini-1.5_Pro-128k when evaluating the stimuli in Liu et al.'s (2024)

Is it clear that the bag on the left seems recycled and the one on the right a plastic bag? I need these stimuli for running an experiment



It's clear that the bag on the right is a plastic bag. However, it's not clear that the bag on the left is recycled.

Here's why:

- **The bag on the left is made of paper, not plastic.** It has a paper-like texture and folds like paper. The green markings and the Chinese characters "环保材料" translate to "environmentally friendly material" which suggests it's intended as a recyclable or environmentally friendly option.
- **The bag on the right is clearly a plastic bag.** Its thin, transparent appearance is typical of plastic bags.

For your experiment, you'll need to ensure the stimuli clearly reflect what you want to test.

- **If you want to compare a recycled paper bag to a plastic bag:** You'll need a clearer image of a paper bag specifically labeled as recycled. While the bag on the left is likely recyclable, it's not explicitly stated.
- **If you want to compare a paper bag to a plastic bag:** The image on the left is suitable, as it's clearly a paper bag. The image on the right is also suitable as it's clearly a plastic bag.

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